# NETWORK TRAFFIC ANALYSIS USING WIRESHARK

1. OBJECTIVE
   To capture and analyze network traffic using Wireshark and understand how DNS resolves domain names and how client-server communication occurs.

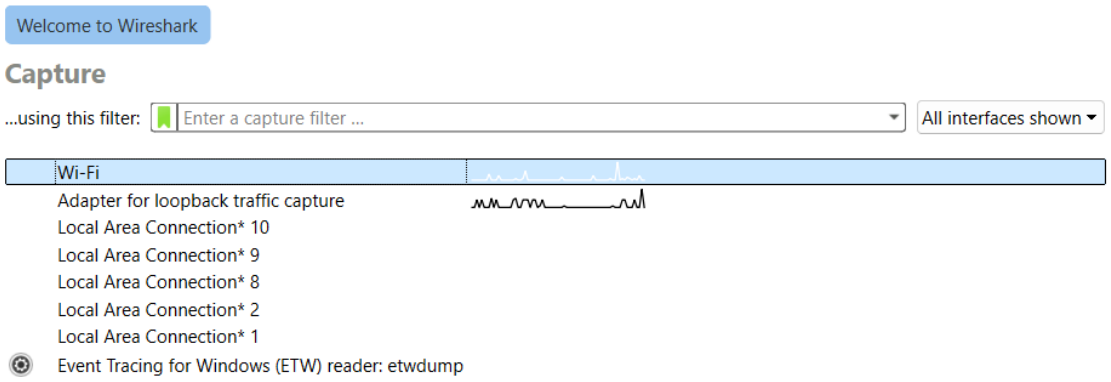
2. TOOLS USED
   Wireshark

3. METHODOLOGY

- Installed and launched Wireshark
- Selected WiFi interface for capturing traffic
- Started packet capture
- Generated traffic by browsing websites
- Stopped capture after a short duration
- Applied DNS and TLS filters for analysis

4. ANALYSIS

4.1. Packet Capture
Network traffic was captured using the WiFi interface. Multiple packets were observed representing real-time communication between the system and external servers.



| No. | Time | Source | Destination | Protocol | Lengtl | Info |
|---|---|---|---|---|---|---|
| 34 | 3.177563700 | 2404:6800:4009:824:… | 2409:40c2:1012:5c53… | QUIC | 84 | Protected Payload (KP0) |
| 35 | 3.177961800 | 2409:40c2:1012:5c53… | 2404:6800:4009:824:… | QUIC | 93 | Protected Payload (KP0), DCID |
| 36 | 3.511205900 | 2404:6800:4009:824:… | 2409:40c2:1012:5c53… | QUIC | 430 | Protected Payload (KP0) |
| 37 | 3.511205900 | 2404:6800:4009:824:… | 2409:40c2:1012:5c53… | QUIC | 83 | Protected Payload (KP0) |
| 38 | 3.511344300 | 2404:6800:4009:824:… | 2409:40c2:1012:5c53… | QUIC | 430 | Protected Payload (KP0) |
| 39 | 3.511742700 | 2409:40c2:1012:5c53… | 2404:6800:4009:824:… | QUIC | 97 | Protected Payload (KP0), DCID |
| 40 | 3.511855200 | 2409:40c2:1012:5c53… | 2404:6800:4009:824:… | QUIC | 95 | Protected Payload (KP0), DCID |
| 41 | 3.514531700 | 2409:40c2:1012:5c53… | 2600:1901:0:47fc:: | TCP | 74 | 19259 → 443 [FIN, ACK] Seq=1 |
| 42 | 3.548873200 | 2404:6800:4009:824:… | 2409:40c2:1012:5c53… | QUIC | 85 | Protected Payload (KP0) |
| 43 | 3.548873200 | 2600:1901:0:47fc:: | 2409:40c2:1012:5c53… | TCP | 74 | 443 → 19259 [FIN, ACK] Seq=1 |
| 44 | 3.549039800 | 2409:40c2:1012:5c53… | 2600:1901:0:47fc:: | TCP | 74 | 19259 → 443 [ACK] Seq=2 Ack=2 |
| 45 | 4.842415700 | 54.169.5.36 | 10.180.202.35 | TCP | 56 | 80 → 32099 [ACK] Seq=1 Ack=1 |
| 46 | 4.842508700 | 10.180.202.35 | 54.169.5.36 | TCP | 54 | [TCP ACKed unseen segment] 32 |
| 47 | 7.608879700 | Intel_c2:6c:21 | c6:a7:af:16:ae:4c | ARP | 42 | Who has 10.180.202.174? Tell |
| 48 | 7.615594100 | c6:a7:af:16:ae:4c | Intel_c2:6c:21 | ARP | 42 | 10.180.202.174 is at c6:a7:af |
| 49 | 8.100235700 | 2409:40c2:1012:5c53… | 2409:40c2:1012:5c53… | ICMPv6 | 86 | Neighbor Solicitation for 240 |
| 50 | 8.105755900 | 2409:40c2:1012:5c53… | 2409:40c2:1012:5c53… | ICMPv6 | 78 | Neighbor Advertisement 2409:4 |
| 51 | 8.323944700 | fe80::c4a7:afff:fe1… | 2409:40c2:1012:5c53… | ICMPv6 | 86 | Neighbor Solicitation for 240 |

## 4.2. DNS Analysis

DNS (Domain Name System) packets were analyzed to understand domain resolution. A DNS query packet was observed where the system requested the IP address of a domain (e.g., www.google.com).

The corresponding DNS response packet contained multiple IP addresses, indicating load balancing.

dns

| No. | Time | Source | Destination | Protocol | Lengtl | Info |
|---|---|---|---|---|---|---|
| 4923 | 68.227330700 | 2409:40c2:1012:5c53… | 2409:40c2:1012:5c53… | DNS | 161 | Standard query response 0x8e25 AAAA clients4.google.com CNAME clients.l.google.c |
| 4924 | 68.227426500 | 2409:40c2:1012:5c53… | 2409:40c2:1012:5c53… | DNS | 149 | Standard query response 0x6b06 A clients4.google.com CNAME clients.l.google.com |
| 4929 | 68.231873300 | 10.180.202.35 | 10.180.202.174 | DNS | 89 | Standard query 0xb8ae A clientservices.googleapis.com |
| 4930 | 68.232331700 | 10.180.202.35 | 10.180.202.174 | DNS | 89 | Standard query 0xa41d AAAA clientservices.googleapis.com |
| 4931 | 68.235893500 | 10.180.202.174 | 10.180.202.35 | DNS | 117 | Standard query response 0xa41d AAAA clientservices.googleapis.com AAAA 2404:6800 |
| 4932 | 68.255015600 | 2409:40c2:1012:5c53… | 2409:40c2:1012:5c53… | DNS | 109 | Standard query 0xb8ae A clientservices.googleapis.com |
| 4933 | 68.263975900 | 10.180.202.174 | 10.180.202.35 | DNS | 105 | Standard query response 0xb8ae A clientservices.googleapis.com A 142.250.206.163 |
| 4956 | 68.281686800 | 2409:40c2:1012:5c53… | 2409:40c2:1012:5c53… | DNS | 125 | Standard query response 0xb8ae A clientservices.googleapis.com A 142.250.70.67 |
| 5217 | 98.806194100 | 10.180.202.35 | 10.180.202.174 | DNS | 76 | Standard query 0xc9f0 HTTPS web.whatsapp.com |
| 5218 | 98.806654300 | 10.180.202.35 | 10.180.202.174 | DNS | 76 | Standard query 0x9bfe AAAA web.whatsapp.com |
| 5219 | 98.806916000 | 10.180.202.35 | 10.180.202.174 | DNS | 76 | Standard query 0x4fa2 A web.whatsapp.com |
| 5220 | 98.807785300 | 10.180.202.35 | 10.180.202.174 | DNS | 89 | Standard query 0x3b73 HTTPS _5222._https.web.whatsapp.com |
| 5221 | 98.808067800 | 10.180.202.35 | 10.180.202.174 | DNS | 76 | Standard query 0x5679 AAAA web.whatsapp.com |
| 5222 | 98.808325500 | 10.180.202.35 | 10.180.202.174 | DNS | 76 | Standard query 0x1335 A web.whatsapp.com |
| 5223 | 98.838906400 | 10.180.202.174 | 10.180.202.35 | DNS | 141 | Standard query response 0x9bfe AAAA web.whatsapp.com CNAME mmx-ds.cdn.whatsapp.n |

udp.stream eq 13

| No. | Time | Source | Destination | Protocol | Lengtl | Info |
|---|---|---|---|---|---|---|
| 112 | 25.277317900 | 2409:40c2:1012:5c53… | 2409:40c2:1012:5c53… | DNS | 95 | Standard query 0x77e5 A docs.google.com |
| 115 | 25.390447600 | 2409:40c2:1012:5c53… | 2409:40c2:1012:5c53… | DNS | 111 | Standard query response 0x77e5 A docs.google.com A 142.251.43.14 |

Wireshark · Follow UDP Stream (udp.stream eq 13) · Wi-Fi

```
w...........docs.google.com.....
w...........docs.google.com.............I....+.
```

```
Domain Name System (response)
    Transaction ID: 0x77e5
  > Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 1
    Authority RRs: 0
    Additional RRs: 0
  v Queries
    > docs.google.com: type A, class IN
  v Answers
    v docs.google.com: type A, class IN, addr 142.251.43.14
        Name: docs.google.com
        Type: A (1) (Host Address)
        Class: IN (0x0001)
        Time to live: 73 (1 minute, 13 seconds)
        Data length: 4
        Address: 142.251.43.14
    [Request In: 112]
    [Time: 113.129700 milliseconds]
```

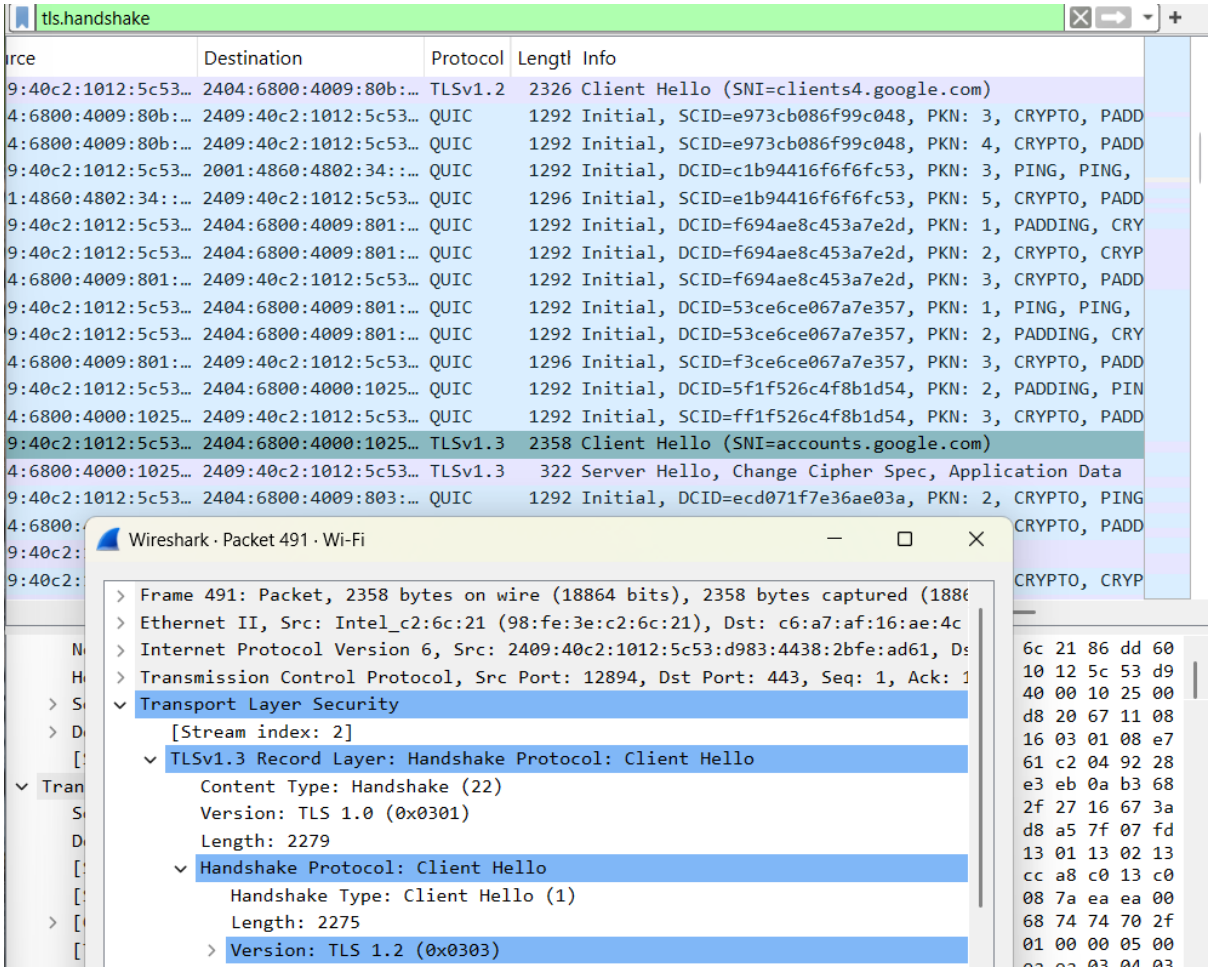
4.3. TLS/HTTPS Analysis –

TLS (Transport Layer Security) packets were analyzed to observe secure communication. A Client Hello message was identified, which initiates the secure connection between the client and server.

The presence of TLS packets indicates that most modern web communication is encrypted.

tls

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 3 | 2.384973100 | 2606:4700:8ca1:d8b7… | 2409:40c2:1012:5c53… | TLSv1.2 | 98 | Application Data |
| 4 | 2.385674200 | 2409:40c2:1012:5c53… | 2606:4700:8ca1:d8b7… | TLSv1.2 | 102 | Application Data |
| 14 | 3.114535000 | 2409:40c2:1012:5c53… | 2404:6800:4009:824:… | QUIC | 1292 | Initial, DCID=ef4727f320b2b49 |
| 16 | 3.114619500 | 2409:40c2:1012:5c53… | 2404:6800:4009:824:… | QUIC | 1292 | Initial, DCID=ef4727f320b2b49 |
| 25 | 3.147222800 | 2404:6800:4009:824:… | 2409:40c2:1012:5c53… | QUIC | 1292 | Initial, SCID=ef4727f320b2b49 |
| 66 | 20.711225000 | 2409:40c2:1012:5c53… | 2606:4700:8d72:7cbc… | QUIC | 1292 | Initial, DCID=9b73929f25c4131 |
| 71 | 20.745497300 | 2606:4700:8d72:7cbc… | 2409:40c2:1012:5c53… | QUIC | 1262 | Initial, SCID=01595baaccc286a |
| 72 | 20.745811500 | 2606:4700:8d72:7cbc… | 2409:40c2:1012:5c53… | QUIC | 1262 | Protected Payload (KP0) |
| 106 | 22.455526000 | 2606:4700:8ca1:d8b7… | 2409:40c2:1012:5c53… | TLSv1.2 | 98 | Application Data |
| 107 | 22.455932100 | 2409:40c2:1012:5c53… | 2606:4700:8ca1:d8b7… | TLSv1.2 | 102 | Application Data |
| 120 | 25.424520200 | 2409:40c2:1012:5c53… | 2404:6800:4009:816:… | TLSv1.3 | 1863 | Client Hello (SNI=docs.google |
| 123 | 25.460938500 | 2404:6800:4009:816:… | 2409:40c2:1012:5c53… | TLSv1.3 | 1298 | Server Hello |
| 124 | 25.461135200 | 2404:6800:4009:816:… | 2409:40c2:1012:5c53… | TLSv1.3 | 1298 | Change Cipher Spec |
| 132 | 25.462079200 | 2404:6800:4009:816:… | 2409:40c2:1012:5c53… | TLSv1.3 | 638 | Application Data |
| 133 | 25.467496700 | 2409:40c2:1012:5c53… | 2404:6800:4009:816:… | TLSv1.3 | 148 | Change Cipher Spec, Applicati |




### 4.4. Packet details

Packet details were analyzed by examining individual packets in Wireshark. The Internet Protocol section provided the source and destination IP addresses, which represent the sender and receiver of the data.

The Transmission Control Protocol (TCP) layer displayed port numbers, where port 443 was identified as the destination port, indicating secure HTTPS communication. These details helped in understanding how data flows between the client and server in a network.

```
> Frame 107: Packet, 102 bytes on wire (816 bits), 102 bytes captured (816 bits)
> Ethernet II, Src: Intel_c2:6c:21 (98:fe:3e:c2:6c:21), Dst: c6:a7:af:16:ae:4c (c
v Internet Protocol Version 6, Src: 2409:40c2:1012:5c53:d983:4438:2bfe:ad61, Dst:
    0110 .... = Version: 6
  > .... 0000 0000 .... .... .... .... .... = Traffic Class: 0x00 (DSCP: CS0, EC
    .... 0001 1100 1110 1000 0100 = Flow Label: 0x1ce84
    Payload Length: 48
    Next Header: TCP (6)
    Hop Limit: 63
  > Source Address: 2409:40c2:1012:5c53:d983:4438:2bfe:ad61
  > Destination Address: 2606:4700:8ca1:d8b7:9a59:4c2:ccf5:ac50
    [Stream index: 1]
> Transmission Control Protocol, Src Port: 13582, Dst Port: 443, Seq: 29, Ack: 49
> Transport Layer Security
```

```
v Transmission Control Protocol, Src Port: 13582, Dst Port: 443, Seq: 29, Ack:
    Source Port: 13582
    Destination Port: 443
    [Stream index: 1]
    [Stream Packet Number: 5]
  > [Conversation completeness: Incomplete (28)]
    [TCP Segment Len: 28]
```

5. <u>OBSERVATIONS-</u>
    - DNS resolves domain names into IP addresses
    - DNS responses may contain multiple IPs
    - TLS is used to encrypt communication
    - Packets contain source IP, destination IP, and protocol details
    - Port 443 is commonly used for secure communication

6. <u>CONCLUSION-</u>
   This project helped in understanding how network communication works in real-world scenarios. It demonstrated how DNS translates domain names into IP addresses and how secure communication is established using TLS.